

Pharmaceutical Formulation and Drug Delivery

Science

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Pharmaceutical Formulation and Drug Delivery (A09676)

Short Title: Pharm Form & Drug Delivery
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author BWHELAN
Credits 5

Level: Advanced

Description of Module / Aims

The module will cover the main technologies related to finished product pharmaceutical formulation and manufacture. Strategies for formulation will be included. Students will carry out an integrated set of experiments from preparation, through formulation, to production and analysis of pharmaceutical products. Work will include formulation, blending, compression, physical testing and analysis.

Programmes

PHAR-0006	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 2 / M
PHAR-0006	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 8 / M

Indicative Content

- Review of pharmaceutical dosage forms: tablets, capsules, injectables, aerosols, creams, ointments and emulsions, suspensions and solutions
- Overview of pharmaceutical development pathway: pre-clinical, clinical and pharmaceutical
- Development pharmaceuticals: pre-formulation studies purpose and implementation, role of excipients; drug properties: physical, chemical, powder flow determination and factors affecting powder flow, polymorphism
- Aerosols: formulation, excipients used, pressurised spraying devices
- Transdermal delivery systems (TDSS): types of TDSS (single layer, multilayer, matrix system), functions of excipients used, compatibility studies, adhesion; dissolution testing required, uniformity/ dissolution testing
- Development, manufacture and testing of pharmaceutical products; unit processes in formulation including mixing, blending and granulation, tableting
- Types of modified release oral dosage forms, disintegrating and non-disintegrating types: manufacture and testing, dissolution testing/bioavailability
- Stability testing of a product: long-term testing, accelerated testing, maintenance of its properties throughout the shelf life; assignment of a shelf life; testing to ICH guidelines
- Development and manufacture of pharmaceutical suspensions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the principles of pharmaceutical development, manufacture for a range of dosage forms including solid dose, suspensions semi-solids and aerosols.
2. Prepare a protocol for the development pharmaceuticals of a finished dosage form, based upon the properties of the active pharmaceutical ingredient and the required bioavailability.
3. Propose and justify a simple formulation and method of manufacture demonstrating a firm understanding of the principle of formulation.
4. Appraise the development and manufacture of immediate and modified release products oral dosage forms, suspensions, aerosols and the rationale for their use.
5. Prepare and implement a protocol for the stability testing of a pharmaceutical product, utilising a reduced testing regiment where appropriate.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	40%	1,4
<i>Assignment</i>	10%	5
<i>Practical</i>	50%	2,3

Assessment Criteria

- <40%:** Little or no demonstration of an understanding of the fundamentals of pharmaceutical formulation and drug delivery. Lack of competence with regard to associated laboratory skills, practices and instrumentation.
- 40%-49%:** Will be able to show limited understanding of the fundamentals of pharmaceutical formulation and drug delivery. Will display some ability with regard to associated laboratory skills, practices and instrumentation.
- 50%-59%:** As well as the above, the student will be to show reasonable understanding of the fundamentals of pharmaceutical formulation and drug delivery. Will display reasonable competence with regard to associated laboratory skills, practices and instrumentation.
- 60%-69%:** In addition, the student will demonstrate more detailed knowledge of pharmaceutical formulation and drug delivery and show ability to analyse problems in this area. Will display very good competence with regard to associated laboratory skills, practices and instrumentation.
- 70%-100%:** All previous to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of pharmaceutical formulation and drug delivery, excellent reasoning skills and well thought out arguments. The learner will also display advanced competence and independence with regard to associated laboratory skills, practices and instrumentation.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	27	
Field Trips	9	
Independent Learning	75	

Essential Material

- Allen, L.V, N. G Popovich and H.C Ansel. _Pharmaceutical Dosage forms and Drug Delivery Systems_. 9th ed. London: Lippincott, Williams & Wilkins, 2011.

Supplementary Material

- Griffin, J.P., J. Posner and G.R. Barker. _The textbook of Pharmaceutical_ _Medicine_. London: Wiley Blackhall, 2013.
- Jithan, A. _Oral Drug Delivery_. India: Pharma book syndicate , 2007.
- Zheng, J. _Formulation and Analytical development for low dose oral drug products_. New Jersey: John Wiley, 2009.

Requested Resources

- Room Type: Chemistry Lab